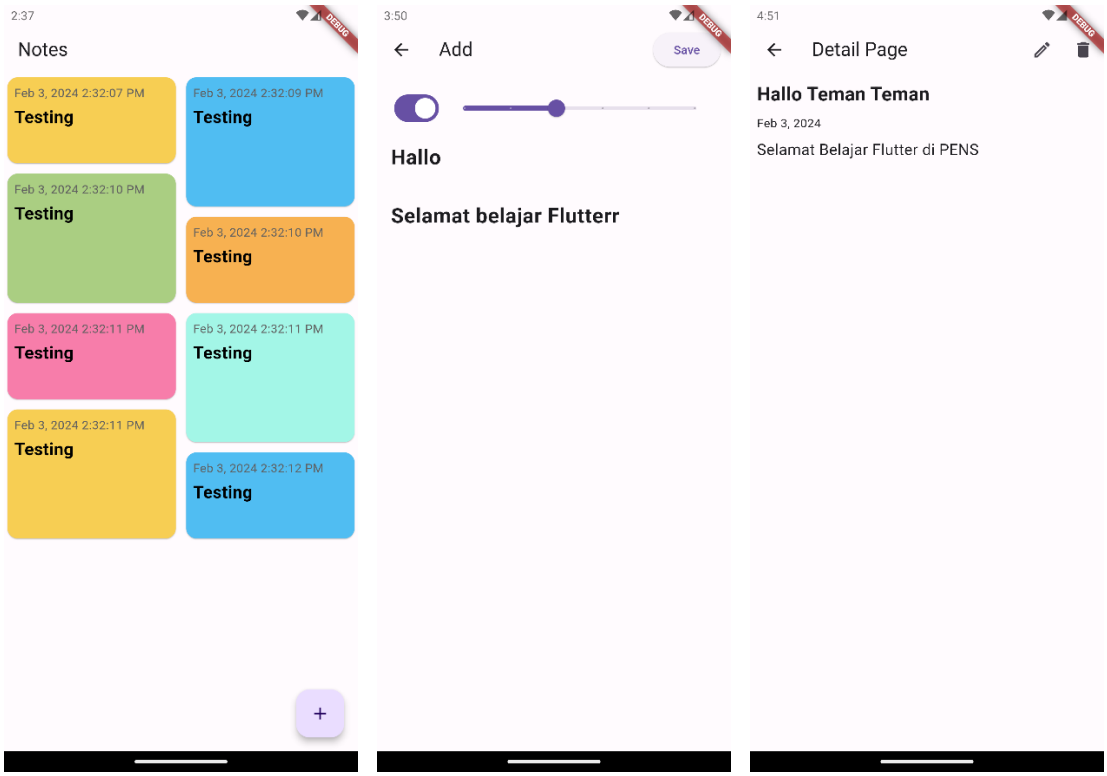
Note App – Database



Inisialisasi

- Buat project baru (ctrl+shift+p) (Empty Project)
- lakukan install dependency **sqflite, intl, flutter_staggered_grid_view** di **pubspec.yaml**

```yaml
name: note_app
description: "A new Flutter project."
publish_to: 'none'
version: 0.1.0

environment:
  sdk: '>=3.2.0 <4.0.0'

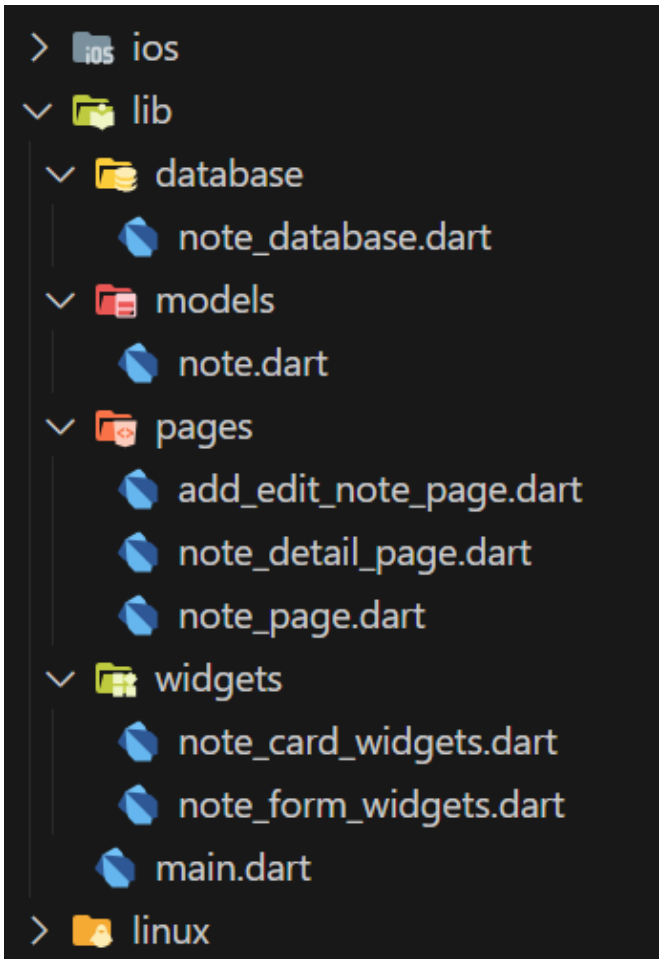
dependencies:
  flutter:
    sdk: flutter
  flutter_staggered_grid_view: ^0.7.0
  intl: ^0.18.1
  sqflite: ^2.3.0

dev_dependencies:
  flutter_test:
    sdk: flutter
```

```
  flutter_lints: ^2.0.0

flutter:
  uses-material-design: true
```

- Buat folder **database, models, pages, widgets** ke dalam folder **lib**
- Silahkan membuat file berikut ini



Menyiapkan Database

- Pada bagian **database /note_database.dart** ketik:

```
import 'package:sqflite/sqflite.dart';


class NoteDatabase {
  static final NoteDatabase instance = NoteDatabase._init();
  NoteDatabase._init();

  static Database? _database;

  Future<Database> get database async {
    if (_database != null) return _database!;
    _database = await _initDB('notes.db');
    return _database!;
  }
}
```

- Pada bagian **database /note_database.dart** lakukan import **path.dart**

```
import 'package:sqflite/sqflite.dart';
import 'package:path/path.dart';
```

- pada bagian **database /note_database.dart** ketik:

```
import 'package:sqflite/sqflite.dart';
import 'package:path/path.dart';

class NoteDatabase {
  static final NoteDatabase instance = NoteDatabase._init();
  NoteDatabase._init();

  static Database? _database;

  Future<Database> get database async {
    if (_database != null) return _database!;
    _database = await _initDB('notes.db');
    return _database!;
  }

  Future<Database> _initDB(String filePath) async {
    final dbPath = await getDatabasesPath();
    final path = join(dbPath, filePath);
```

```
    return await openDatabase(path, version: 1, onCreate: _createDB);
  }
}
```

- Selanjutnya karena masih belum ada database maka kita buatkan pada bagian **models/note.dart**

```
const String tableNotes = "notes";

class NoteFields {
  static final List<String> values = [];

  static const String id = "_id";
  static const String isImportant = "isImportant";
  static const String number = "number";
  static const String title = "title";
  static const String description = "description";
  static const String time = "time";
}
```

- pada bagian **database /note_database.dart** ketik:

```
import 'package:note_app/models/note.dart';
```

- pada bagian **database /note_database.dart** ketik:

```
  Future _createDB(Database db, int version) async {
    const sql = '''
CREATE TABLE $tableNotes(
       ${NoteFields.id} INTEGER PRIMARY KEY AUTOINCREMENT,
       ${NoteFields.isImportant} BOOLEAN NOT NULL,
       ${NoteFields.number} INTEGER NOT NULL,
       ${NoteFields.title} TEXT NOT NULL,
       ${NoteFields.description} TEXT NOT NULL,
       ${NoteFields.time} TEXT NOT NULL
     )
''';
    await db.execute(sql);
  }
```

- pada bagian **models/note.dart** ketik:

```
class Note {
  final int? id;
  final bool isImportant;
  final int number;
  final String title;
  final String description;
  final DateTime createdTime;


  Note(
      {this.id,
      required this.isImportant,
      required this.number,
      required this.title,
      required this.description,
      required this.createdTime});
}
```

- Tambahkan dibawah **Note** pada bagian **models/note.dart**:

```
  Note(
      {this.id,
      required this.isImportant,
      required this.number,
      required this.title,
      required this.description,
      required this.createdTime});

  Note copy(
      {int? id,
      bool? isImportant,
      int? number,
      String? title,
      String? description,
      DateTime? createdTime}) {
    return Note(
        id: id ?? this.id,
        isImportant: isImportant ?? this.isImportant,
        number: number ?? this.number,
        title: title ?? this.title,
        description: description ?? this.description,
        createdTime: createdTime ?? this.createdTime);
  }
```

```
  static Note fromJson(Map<String, Object?> json) {
    return Note(
        id: json[NoteFields.id] as int?,
        isImportant: json[NoteFields.isImportant] == 1,
        number: json[NoteFields.number] as int,
        title: json[NoteFields.title] as String,
        description: json[NoteFields.description] as String,
        createdTime: DateTime.parse(json[NoteFields.time] as String));
  }

  Map<String, Object?> toJson() => {
        NoteFields.id: id,
        NoteFields.isImportant: isImportant ? 1 : 0,
        NoteFields.number: number,
        NoteFields.title: title,
        NoteFields.description: description,
        NoteFields.time: createdTime.toIso8601String()
      };
}
```

- Tambahkan dibawah **CRUD** pada **database/note_database.dart** ketik:

```
  // CRUD
  Future<Note> create(Note note) async {
    final db = await instance.database;
    final id = await db.insert(tableNotes, note.toJson());
    return note.copy(id: id);
  }

  Future<List<Note>> getAllNotes() async {
    final db = await instance.database;
    final result = await db.query(tableNotes);
    return result.map((json) => Note.fromJson(json)).toList();
  }
```

- Pada bagian **pages/note_page.dart** ketik:

```
import 'package:flutter/material.dart';

class NotePage extends StatefulWidget {
  const NotePage({super.key});

  @override
  State<NotePage> createState() => _NotePageState();
}
```

```dart
class _NotePageState extends State<NotePage> {
  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Tambahkan dibawah **_NotePageState** pada **pages/note_page.dart** ketik:

```dart
class _NotePageState extends State<NotePage> {
  late List<Note> notes;
  var isLoading = false;

  Future refreshNotes() async {
    setState(() {
      isLoading = true;
    });

    notes = await NoteDatabase.instance.getAllNotes();

    setState(() {
      isLoading = false;
    });
  }

  @override
  void initState() {
    super.initState();
    refreshNotes();
  }

  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Rubah **Placeholder** pada **pages/note_page.dart** ketik:

```
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Notes"),
      ),
      body: isLoading
          ? const Center(
              child: CircularProgressIndicator(),
            )
          : notes.isEmpty
              ? const Text("Notes Kosong")
              : MasonryGridView.count(
                  crossAxisCount: 2,
                  itemCount: notes.length,
                  mainAxisSpacing: 4,
                  crossAxisSpacing: 4,
                  itemBuilder: (context, index) {
                    return Text("data");
                  }),
    );
  }
```

- Pada bagian **pages/note_page.dart** lakukan import package yang diperlukan

```
import 'package:flutter/material.dart';
import 'package:flutter_staggered_grid_view/flutter_staggered_grid_view.dart';
import 'package:note_app/database/note_database.dart';
import 'package:note_app/models/note.dart';
```

- Rubah pada **main.dart** ketik:

```
import 'package:flutter/material.dart';
import 'package:note_app/pages/note_page.dart';

void main() {
  runApp(const MainApp());
}

class MainApp extends StatelessWidget {
  const MainApp({super.key});

  @override
```

```
  Widget build(BuildContext context) {
    return const MaterialApp(
      home: NotePage(),
    );
  }
}
```

- Hasil akhir yang ditunjukan sebagai berikut



Membuat Note Card

- Silahkan membuat **StatelessWidget** terlebih dahulu di **widgets/note_card_widgets.dart**

```
import 'package:flutter/material.dart';

class NoteCardWidget extends StatelessWidget {
  const NoteCardWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Buat list warna yang akan dipakai oleh note pada **widgets/note_card_widgets.dart**

```
import 'package:flutter/material.dart';

final _lightColors = [
  Colors.amber.shade300,
  Colors.lightBlue.shade300,
  Colors.lightGreen.shade300,
  Colors.orange.shade300,
  Colors.pinkAccent.shade100,
  Colors.tealAccent.shade100
];

class NoteCardWidget extends StatelessWidget {
  const NoteCardWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Lakukan akses model data pada **widgets/note_card_widgets.dart**

```
import 'package:flutter/material.dart';
import 'package:note_app/models/note.dart';

final _lightColors = [
```

```
  Colors.amber.shade300,
  Colors.lightBlue.shade300,
  Colors.lightGreen.shade300,
  Colors.orange.shade300,
  Colors.pinkAccent.shade100,
  Colors.tealAccent.shade100
];

class NoteCardWidget extends StatelessWidget {
  final Note note;
  final int index;
  const NoteCardWidget({super.key, required this.note, required this.index});

  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Lakukan perubahan pada **@override** di **widgets/note_card_widgets.dart**

```
import 'package:flutter/material.dart';
import 'package:intl/intl.dart';
import 'package:note_app/models/note.dart';

...

  @override
  Widget build(BuildContext context) {
    final time = DateFormat.yMMMd().add_jms().format(note.createdTime);
    final minHeight = getMinHeight(index);
    final color = _lightColors[index % _lightColors.length];
    return Card(
      color: color,
      child: Container(
        constraints: BoxConstraints(minHeight: minHeight),
        padding: const EdgeInsets.all(8),
        child: Column(
          mainAxisSize: MainAxisSize.min,
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            Text(
              time,
              style: TextStyle(color: Colors.grey.shade700),
```

```
          ),
          const SizedBox(
            height: 4,
          ),
          Text(
            note.title,
            style: const TextStyle(
              color: Colors.black,
              fontSize: 20,
              fontWeight: FontWeight.bold,
            ),
          ),
        ],
      ),
    ),
  );
}
```

- Tambahkan **getMinHeight** dibagian paling bawah di **widgets/note_card_widgets.dart**

```
double getMinHeight(int index) {
  switch (index % 4) {
    case 0:
      return 100;
    case 1:
      return 150;
    case 2:
      return 150;
    case 3:
      return 100;
    default:
      return 100;
  }
}
```

- Pada **pages/note_page.dart** Tambahkan kode berikut di dalam MasonryGridView

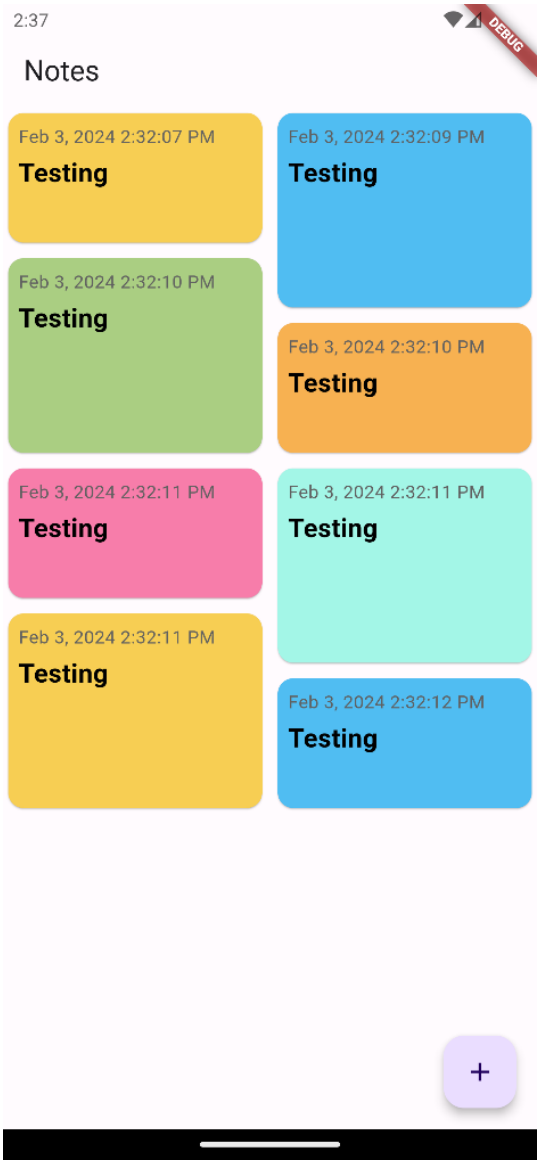
```
        : notes.isEmpty
            ? const Text("Notes Kosong")
            : MasonryGridView.count(
                crossAxisCount: 2,
                itemCount: notes.length,
                mainAxisSpacing: 4,
                crossAxisSpacing: 4,
                itemBuilder: (context, index) {
```

```
                final note = notes[index];
                return GestureDetector(
                    child: NoteCardWidget(note: note, index: index));
              }),
```

- Agar dapat menambahkan note maka perlu dibuatkan tombol. Tambahkan **floatingActionButton** berikut Pada **pages/note_page.dart** dibagian setelah **MasonryGridView**

```
          : MasonryGridView.count(
              crossAxisCount: 2,
              itemCount: notes.length,
              mainAxisSpacing: 4,
              crossAxisSpacing: 4,
              itemBuilder: (context, index) {
                final note = notes[index];
                return GestureDetector(
                    child: NoteCardWidget(note: note, index: index));
              }),
  floatingActionButton: FloatingActionButton(
    onPressed: () async {
      final note = Note(
          isImportant: false,
          number: 1,
          title: "Testing",
          description: "Desc testing",
          createdTime: DateTime.now());
      await NoteDatabase.instance.create(note);
      refreshNotes();
    },
    child: const Icon(Icons.add),
  ),
```

- Hasil akhir yang ditunjukan sebagai berikut



Menambahkan Menu Add Note

- Silahkan membuat **StatefulWidget** terlebih dahulu di **pages/ add_edit_note_page.dart**

```
import 'package:flutter/material.dart';

class AddEditNotePage extends StatefulWidget {
  const AddEditNotePage({super.key});

  @override
  State<AddEditNotePage> createState() => _AddEditNotePageState();
}

class _AddEditNotePageState extends State<AddEditNotePage> {
  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Pada bagian **pages/ add_edit_note_page.dart** ketik:

```
import 'package:flutter/material.dart';

class AddEditNotePage extends StatefulWidget {
  const AddEditNotePage({super.key});

  @override
  State<AddEditNotePage> createState() => _AddEditNotePageState();
}

class _AddEditNotePageState extends State<AddEditNotePage> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Add"),
      ),
      body: Form(
        child: Container(),
      ),
    );
  }
}
```

- Rubah **floatingActionButton** pada **pages/note_page.dart** ketik:

```
import 'package:flutter/material.dart';
import 'package:flutter_staggered_grid_view/flutter_staggered_grid_view.dart';
import 'package:note_app/database/note_database.dart';
import 'package:note_app/models/note.dart';
import 'package:note_app/pages/add_edit_note_page.dart';
import 'package:note_app/widgets/note_card_widgets.dart';


...

      floatingActionButton: FloatingActionButton(
        onPressed: () async {
          // final note = Note(
          //     isImportant: false,
          //     number: 1,
          //     title: "Testing",
          //     description: "Desc testing",
          //     createdTime: DateTime.now());
          // await NoteDatabase.instance.create(note);
          await Navigator.push(context,
              MaterialPageRoute(builder: (context) => AddEditNotePage()));

          refreshNotes();
        },
        child: const Icon(Icons.add),
      ),
```

- Hasil yang ditunjukan sebagai berikut



Menambahkan Form Add Note

- Rubah **AddEditNotePageStat** pada bagian **pages/ add_edit_note_page.dart** ketik:

```
class AddEditNotePage extends StatefulWidget {
  final Note? note;
  const AddEditNotePage({super.key, this.note});

  @override
  State<AddEditNotePage> createState() => _AddEditNotePageState();
}

class _AddEditNotePageState extends State<AddEditNotePage> {
  late bool isImportant;
  late int number;
  late String title;
  late String description;
  final _formKey = GlobalKey<FormState>();
  var isUpdateForm = false;

  @override
  void initState() {
    super.initState();
    isImportant = widget.note?.isImportant ?? false;
    number = widget.note?.number ?? 0;
    title = widget.note?.title ?? "";
    description = widget.note?.description ?? "";
    isUpdateForm = widget.note != null;
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

- Rubah **Widget build** pada bagian **pages/ add_edit_note_page.dart**

```
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Add"),
        actions: [buildButtonSave()],
      ),
      body: Form(
        key: _formKey,
```

```
        child: Container(),
      ),
    );
  }

  buildButtonSave() {
    return Padding(
      padding: const EdgeInsets.symmetric(vertical: 8, horizontal: 12),
      child: ElevatedButton(
        onPressed: () async {},
        child: const Text("Save"),
      ),
    );
  }
}
```

- Maka akan muncul tombol save seperti yang ditunjukan sebagai berikut



- Silahkan membuat **StatelessWidget** terlebih dahulu di **widgets/ note_form_widgets.dart**

```
import 'package:flutter/material.dart';

class NoteFormWidget extends StatelessWidget {
  const NoteFormWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Selanjutnya siapkan **field** yang diperlukan untuk **form** pada bagian **widgets/ note_form_widgets.dart**

```
import 'package:flutter/material.dart';

class NoteFormWidget extends StatelessWidget {
  const NoteFormWidget(
      {Key? key,
      required this.isImportant,
      required this.number,
      required this.title,
      required this.description,
      required this.onChangeIsImportant,
      required this.onChangeNumber,
      required this.onChangeTitle,
      required this.onChangeDescription})
      : super(key: key);

  final bool isImportant;
  final int number;
  final String title;
  final String description;
  final ValueChanged<bool> onChangeIsImportant;
  final ValueChanged<int> onChangeNumber;
  final ValueChanged<String> onChangeTitle;
  final ValueChanged<String> onChangeDescription;

  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
```

```
}
```

- Lakukan perubahan pada **@override Widget Build** di bagian **widgets/ note_form_widgets.dart**

```
  @override
  Widget build(BuildContext context) {
    return SingleChildScrollView(
      child: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(children: [
          Row(
            children: [
              Switch(value: isImportant, onChanged: onChangeIsImportant),
              Expanded(
                  child: Slider(
                value: number.toDouble(),
                min: 0,
                max: 5,
                divisions: 5,
                onChanged: (value) => onChangeNumber(value.toInt()),
              ))
            ],
          ),
          buildTitleField(),
          const SizedBox(
            height: 8,
          ),
          buildDescriptionField()
        ]),
      ),
    );
  }
```

- Tambahkan **buildTitleField** dan **buildDescriptionField** di bagian paling bawah setelah **@override Widget Build** pada file **widgets/ note_form_widgets.dart**

```
  buildTitleField() {
    return TextFormField(
      maxLines: 1,
      initialValue: title,
      style: const TextStyle(
        // color: Colors.white70,
        fontWeight: FontWeight.bold,
```

```
        fontSize: 24,
      ),
      decoration: const InputDecoration(
          border: InputBorder.none,
          hintText: "Title",
          hintStyle: TextStyle(color: Colors.grey)),
      validator: (title) {
        return title != null && title.isEmpty
            ? 'The title cannot be empty'
            : null;
      },
      onChanged: onChangeTitle,
    );
  }

  buildDescriptionField() {
    return TextFormField(
      maxLines: null,
      initialValue: description,
      style: const TextStyle(
        // color: Colors.white70,
        fontWeight: FontWeight.bold,
        fontSize: 24,
      ),
      decoration: const InputDecoration(
          border: InputBorder.none,
          hintText: "Type something...",
          hintStyle: TextStyle(color: Colors.grey)),
      validator: (desc) {
        return desc != null && desc.isEmpty ? 'The desc cannot be empty' : null;
      },
      onChanged: onChangeDescription,
    );
  }
```

- Persiapkan **function** pada **_formKey** di bagian **pages/ add_edit_note_page.dart**

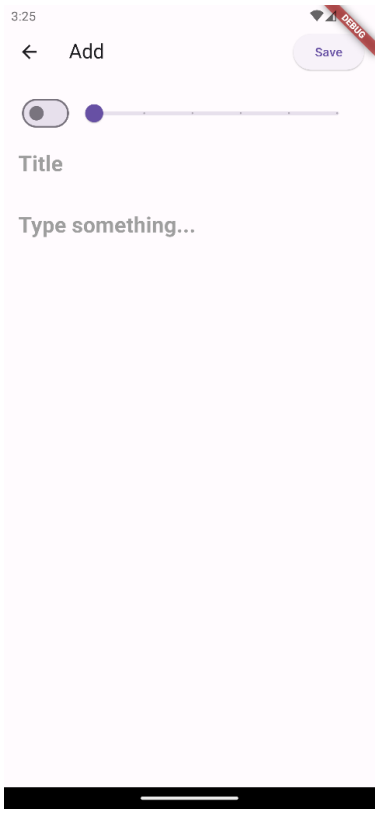
```
import 'package:flutter/material.dart';
import 'package:note_app/widgets/note_form_widgets.dart';

...

 @override
  Widget build(BuildContext context) {
```

```
  return Scaffold(
    appBar: AppBar(
      title: Text("Add"),
      actions: [buildButtonSave()],
    ),
    body: Form(
      key: _formKey,
      child: NoteFormWidget(
        isImportant: isImportant,
        number: number,
        title: title,
        description: description,
        onChangeIsImportant: (value) {},
        onChangeNumber: (value) {},
        onChangeTitle: (value) {},
        onChangeDescription: (value) {},
      ),
    ),
  );
}
```

- Maka akan muncul form sebagai berikut



Menyimpan Input Form Add Note

- Rubah **function** pada **_formKey** di bagian **pages/ add_edit_note_page.dart**

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text("Add"),
      actions: [buildButtonSave()],
    ),
    body: Form(
      key: _formKey,
      child: NoteFormWidget(
        isImportant: isImportant,
        number: number,
        title: title,
        description: description,
        onChangeIsImportant: (value) {
          setState(() {
            isImportant = value;
          });
        },
        onChangeNumber: (value) {
          setState(() {
            number = value;
          });
        },
        onChangeTitle: (value) {
          title = value;
        },
        onChangeDescription: (value) {
          description = value;
        },
      ),
    ),
  );
}
```

- Rubah **buildButtonSave** di bagian **pages/ add_edit_note_page.dart**

```
buildButtonSave() {
  return Padding(
    padding: const EdgeInsets.symmetric(vertical: 8, horizontal: 12),
    child: ElevatedButton(
      onPressed: () async {
        final isValid = _formKey.currentState!.validate();
        if (isValid) {
          // Tambah data
          await addNote();

          // Tutup halaman
          Navigator.pop(context);
        }
      },
      child: const Text("Save"),
    ),
  );
}
```

- Tambahkan method **addNote** setelah **buildButtonSave** di bagian **pages/ add_edit_note_page.dart**

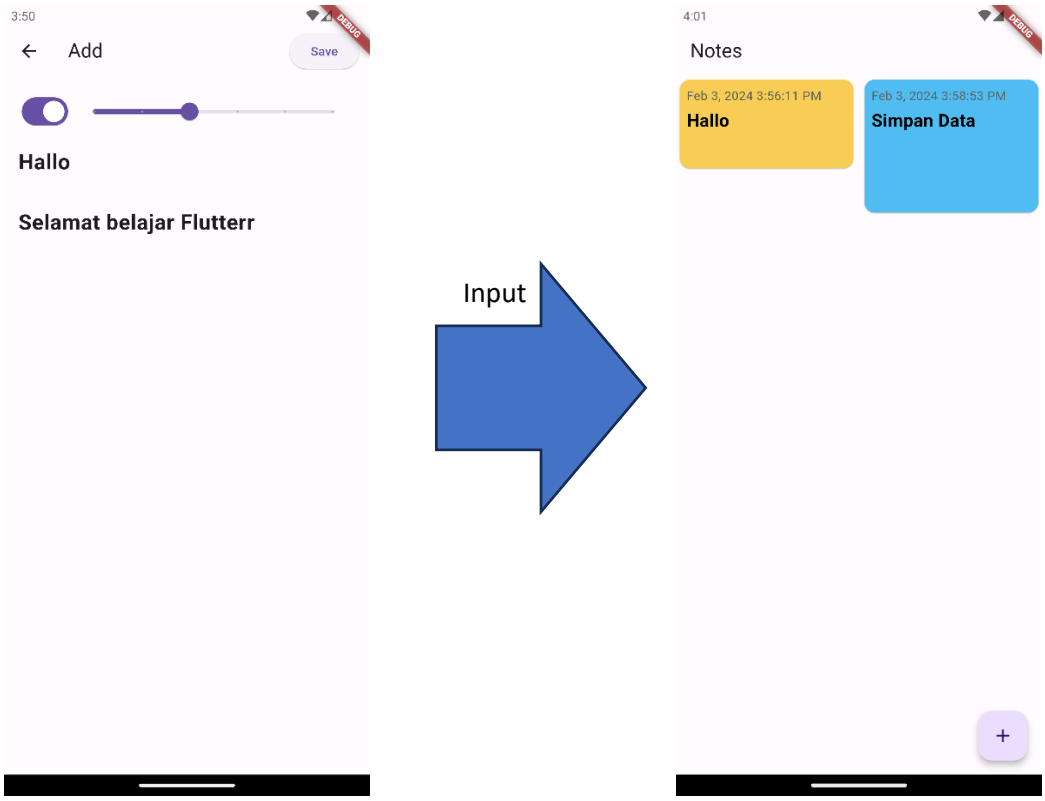
```
buildButtonSave() {
  return Padding(
    padding: const EdgeInsets.symmetric(vertical: 8, horizontal: 12),
    child: ElevatedButton(
      onPressed: () async {
        final isValid = _formKey.currentState!.validate();
        if (isValid) {
          // Tambah data
          await addNote();

          // Tutup halaman
          Navigator.pop(context);
        }
      },
      child: const Text("Save"),
    ),
  );
}

Future addNote() async {
  final note = Note(
```

```
      isImportant: isImportant,
      number: number,
      title: title,
      description: description,
      createdTime: DateTime.now(),
    );
    await NoteDatabase.instance.create(note);
  }
```

- Hasil akhir yang ditunjukan sebagai berikut



Melakukan Update dan Delete Note

- Tambahkan **getNoteById** dibawah **getAllNotes** pada bagian **database /note_database.dart** ketik:

```
Future<List<Note>> getAllNotes() async {
  final db = await instance.database;
  final result = await db.query(tableNotes);
  return result.map((json) => Note.fromJson(json)).toList();
}

Future<Note> getNoteById(int id) async {
  final db = await instance.database;
  final result = await db
      .query(tableNotes, where: '${NoteFields.id} = ?', whereArgs: [id]);
  if (result.isNotEmpty) {
    return Note.fromJson(result.first);
  } else {
    throw Exception('ID $id not found');
  }
}
```

- Tambahkan **deleteNoteById** dan **updateNote** dibawah **getNoteById** pada bagian **database /note_database.dart** ketik:

```
Future<Note> getNoteById(int id) async {
  final db = await instance.database;
  final result = await db
      .query(tableNotes, where: '${NoteFields.id} = ?', whereArgs: [id]);
  if (result.isNotEmpty) {
    return Note.fromJson(result.first);
  } else {
    throw Exception('ID $id not found');
  }
}

Future<int> deleteNoteById(int id) async {
  final db = await instance.database;
  return await db.delete(
    tableNotes,
    where: '${NoteFields.id} = ?',
    whereArgs: [id],
  );
}

Future<int> updateNote(Note note) async {
```

```
  final db = await instance.database;
  return await db.update(
    tableNotes,
    note.toJson(),
    where: '${NoteFields.id} = ?',
    whereArgs: [note.id],
  );
}
```

- Lakukan perubahan pada **buildButtonSave** terlebih dahulu di **pages** / **add_edit_note_page.dart**

```
buildButtonSave() {
  return Padding(
    padding: const EdgeInsets.symmetric(vertical: 8, horizontal: 12),
    child: ElevatedButton(
      onPressed: () async {
        final isValid = _formKey.currentState!.validate();
        if (isValid) {
          if (isUpdateForm) {
            // Update data
            await updateNote();
          } else {
            // Tambah data
            await addNote();
          }
          // Tutup halaman
          Navigator.pop(context);
        }
      },
      child: const Text("Save"),
    ),
  );
}
```

- Lakukan penambahan **updateNote** dibagian paling akhir **pages** / **add_edit_note_page.dart**

```
Future addNote() async {
  final note = Note(
    isImportant: isImportant,
    number: number,
    title: title,
    description: description,
    createdTime: DateTime.now(),
  );
  await NoteDatabase.instance.create(note);
```

```
  }

  Future updateNote() async {
    final updateNote = widget.note?.copy(
      isImportant: isImportant,
      number: number,
      title: title,
      description: description,
      createdTime: DateTime.now(),
    );
    await NoteDatabase.instance.updateNote(updateNote!);
  }
```

- Silahkan membuat **StatefulWidget** terlebih dahulu di **pages /note_detail_page.dart**

```
import 'package:flutter/material.dart';

class NoteDetailPage extends StatefulWidget {
  const NoteDetailPage({super.key});

  @override
  State<NoteDetailPage> createState() => _NoteDetailPageState();
}

class _NoteDetailPageState extends State<NoteDetailPage> {
  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Rubah **NoteDetailPage** pada bagian **pages /note_detail_page.dart**

```
import 'package:flutter/material.dart';

class NoteDetailPage extends StatefulWidget {
  final int id;
  const NoteDetailPage({super.key, required this.id});

  @override
  State<NoteDetailPage> createState() => _NoteDetailPageState();
}

class _NoteDetailPageState extends State<NoteDetailPage> {
```

```
  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
}
```

- Buat Method **refreshNote** dan **initState** didalam **_NoteDetailPageState** pada bagian **pages /note_detail_page.dart**

```
import 'package:flutter/material.dart';
import 'package:note_app/database/note_database.dart';
import 'package:note_app/models/note.dart';

...

class _NoteDetailPageState extends State<NoteDetailPage> {
  late Note note;
  var isLoading = false;

  Future refreshNote() async {
    setState(() {
      isLoading = true;
    });

    note = await NoteDatabase.instance.getNoteById(widget.id);

    setState(() {
      isLoading = false;
    });
  }

  @override
  void initState() {
    super.initState();
    refreshNote();
  }

  @override
  Widget build(BuildContext context) {
    return const Placeholder();
  }
```

- Lakukan perubahan pada **Widget Build** dibagian **Placeholder** di **pages /note_detail_page.dart**

```dart
import 'package:flutter/material.dart';
import 'package:intl/intl.dart';
import 'package:note_app/database/note_database.dart';
import 'package:note_app/models/note.dart';

...

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Detail Page"),
        actions: [editButton(), deleteButton()],
      ),
      body: isLoading
          ? const Center(
              child: CircularProgressIndicator(),
            )
          : ListView(
              padding: const EdgeInsets.all(8),
              children: [
                Text(
                  note.title,
                  style: const TextStyle(
                    // color: Colors.white,
                    fontSize: 22,
                    fontWeight: FontWeight.bold,
                  ),
                ),
                const SizedBox(
                  height: 8,
                ),
                Text(
                  DateFormat.yMMMd().format(note.createdTime),
                  // style: const TextStyle(color: Colors.white38),
                ),
                const SizedBox(
                  height: 8,
                ),
                Text(
                  note.description,
                  style: const TextStyle(
                    // color: Colors.white70,
```

```
                    fontSize: 18,
                  ),
                ),
              ],
            ),
      );
  }
```

- Tambahkan **Widget editButton** dan **deleteButton** dibagian paling akhir pada **pages /note_detail_page.dart**

```
import 'package:flutter/material.dart';
import 'package:intl/intl.dart';
import 'package:note_app/database/note_database.dart';
import 'package:note_app/models/note.dart';
import 'package:note_app/pages/add_edit_note_page.dart';

...

  Widget editButton() => IconButton(
      icon: const Icon(Icons.edit_outlined),
      onPressed: () async {
        if (isLoading) return;
        await Navigator.push(
          context,
          MaterialPageRoute(
            builder: (context) => AddEditNotePage(
              note: note,
            ),
          ),
        );
        refreshNote();
      });

  Widget deleteButton() => IconButton(
        icon: const Icon(Icons.delete),
        onPressed: () async {
          if (isLoading) return;
          await NoteDatabase.instance.deleteNoteById(widget.id);
          Navigator.pop(context);
        },
      );
```

- Lakukan penambahan kode pada **GestureDetector** di **pages/note_page.dart**

```
import 'package:flutter/material.dart';
import 'package:flutter_staggered_grid_view/flutter_staggered_grid_view.dart';
import 'package:note_app/database/note_database.dart';
import 'package:note_app/models/note.dart';
import 'package:note_app/pages/add_edit_note_page.dart';
import 'package:note_app/pages/note_detail_page.dart';
import 'package:note_app/widgets/note_card_widgets.dart';

...

              : MasonryGridView.count(
                  crossAxisCount: 2,
                  itemCount: notes.length,
                  mainAxisSpacing: 4,
                  crossAxisSpacing: 4,
                  itemBuilder: (context, index) {
                    final note = notes[index];
                    return GestureDetector(
                        onTap: () async {
                          await Navigator.push(
                            context,
                            MaterialPageRoute(
                              builder: (context) =>
                                  NoteDetailPage(id: note.id!),
                            ),
                          );
                          refreshNotes();
                        },
                        child: NoteCardWidget(note: note, index: index));
                  }),
      floatingActionButton: FloatingActionButton(
        onPressed: () async {
```

- Hasil akhir yang ditunjukan sebagai berikut

